

Practical Course for Creating Stereoscopic VTuber

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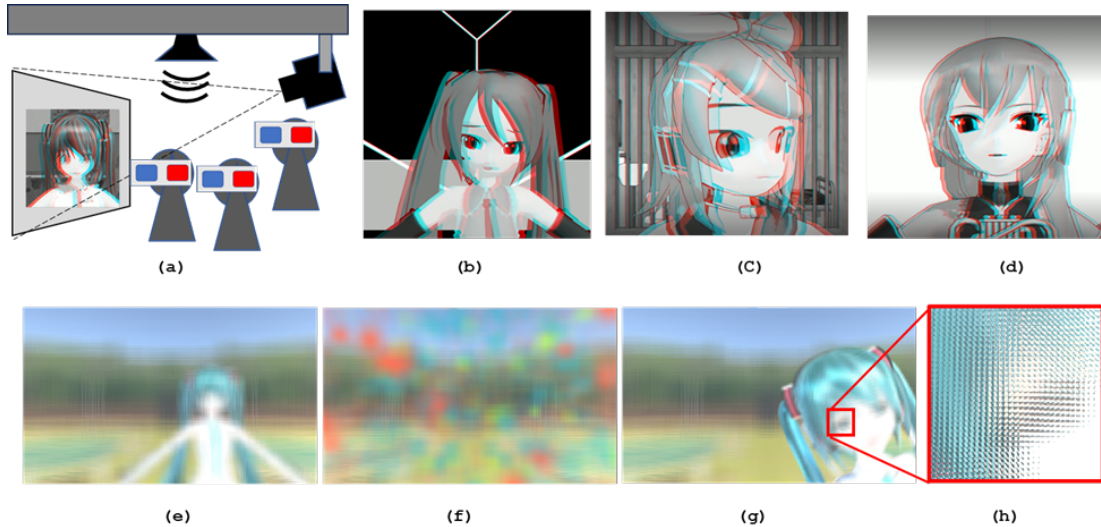


Figure 1: Stereoscopic VTubers. (a) Presentation session (I) using projection of the resultant anaglyph video with audio. (b)–(d) are created anaglyph VTubers by students. the avatars used and the names of the student authors are as follows: (b) “Hatsune Miku (Animasa)” by Takuya Yoshiyama, (c) “Kagamine Rin (Animasa)” by Junya Ishikawa, and (d) “Megurine Luka (Animasa)” by Yuki Imai. (e)–(g) “Menbou-shiki-Hatsune-Miku” featured a combination of an autostereoscopic VTuber with moving background (fireworks) by the author: Kazuya Sasajima at the presentation session (II). (h) Close-up of the fly’s eye lenses for autostereoscopic viewing. These works are depictions of Crypton Future Media’s character “Hatsune Miku”, “Kagamine Rin” and “Megurine Luka” based on the Piapro Character License. [©Crypton Future Media, INC. www.piapro.net. 2023 ; Menbou-shiki-Hatsune-Miku.2022.]

ABSTRACT

The authors conduct a practical class for third-year students in the Information Media Studies Department, focused on creating stereoscopic CG content. They focus on virtual YouTubers (VTubers), that has been gaining popularity on the internet. Despite the current VTubers being created as 3D models, they lack actual stereoscopic capabilities. VTuber characters can feel more lifelike upon introduction of stereoscopic visuals, allowing creators to expand their ideas and unlocking significant potential in the metaverse era. In this study, each student captures video footage of their upper body using their laptop’s web camera. Then, they combine various freely available software tools online to create stereoscopic VTuber videos without the need for special equipment. The students produce content using techniques, such as anaglyph videos (using red-blue

glasses), lenticular lenses, and fly’s eye lenses for autostereoscopic viewing. They also add audio to the VTuber content, and at the end of the exercise, they hold a “Stereoscopic VTuber” presentation, where mutual evaluations are conducted. Results of the student surveys indicate a positive outcome in terms of their increased interest in related technologies and enhanced creative motivation. This practical course holds significant values as an education that combines information media art, system-oriented thinking, and AI technologies.

CCS CONCEPTS

• Social and professional topics; • Professional topics; • Computing education; • Computing education programs; • Computing methodologies; • Computer graphics; • Animation; • Motion capture; • Applied computing; • Arts and humanities; • Media arts;

KEYWORDS

3D, CG, stereoscopic, VTuber

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1 INTRODUCTION

In the Information Media Studies Department, where the authors are affiliated, many students aim to become engineers or creators in fields, such as gaming, CG, video, and sound. As part of this pursuit, the department offers a practical class on creating stereoscopic CG content in the Media Art field [Yanaka and Yamanouchi, 2015].

Traditionally, research on stereoscopy has focused on display methods, but the use of these display methods and devices to applications and contents should be explored to cater to a wide range of needs.

While research on educational stereoscopic visualization has existed in fields, such as medicine and design [D. Kang et al., 2022; Chu and Chien, 2017], studies on appealing stereoscopic educational content in the Media Art domain are limited.

In recent years, virtual YouTubers (VTubers), who appear as CG avatars created by YouTubers themselves, have become popular on the internet as an evolution of a video-sharing platform “YouTube” [Lawrenson, 2022]. Current VTubers, even if created as 3D models, lack stereoscopic capabilities because users typically view them on regular displays. By introducing stereoscopic visuals, VTuber characters can feel more lifelike than flat 2D images, enhancing interactive experiences and significantly expanding the creative expression for content creators. Furthermore, in the near future, with the introduction of the metaverse era, everyone can create their own avatars, holding significant possibilities.

In addition, VTuber technology is a BlackBox technology that combines motion extraction through image processing with CG animation techniques. Therefore, learning VTuber technology allows students to experience a wide range of information technology simultaneously.

However, given that creating original VTuber avatar models deviates from the main purpose of this class, students are encouraged to download and use existing models available on the internet while considering copyright issues.

2 DESIGN OF THE PRACTICAL CLASS

The target audience includes third-year undergraduate students at the engineering university, where the authors are affiliated. A total of 27 students participate in the practical class, with two authors providing guidance as instructors. Students generally own personal laptops equipped with Intel (R) Core (TM) i7-1165G7 @ 2.80GHz + 8MB RAM and a web camera, running Windows 10 64-bit. All video input, processing, and output are accomplished solely using these laptops.

The class was conducted once a week for 14 weeks, with each session lasting 180 minutes. The content is divided into three categories: (A), (B), and (C).

(A) For the first three sessions, a stereoscopic camera (FinePixReal3DW3) [DEGICAME-Watch, 2010] is used as the input device to generate stereoscopic images by chroma key compositing real-world subjects against a background. Stereoscopic viewing is

achieved either through anaglyph videos, which are observed using red–blue glasses on a regular display, or through polarization-based displays (RDT233WX-3D) [Impress Watch, 2011] provided by the instructor; they are observed using polarized glasses.

(B) From the fourth to the seventh session, students create stereoscopic VTuber videos featuring CG avatars with the motion of their upper bodies. These videos are viewed in monochrome anaglyph format. Anaglyph videos are produced by merging two parallax videos into red–blue parallax videos, which are then observed using red–blue glasses. This approach offers advantages, such as cost-effectiveness, clear stereoscopic viewing on regular displays, and the ability for multiple learners to watch simultaneously. However, it has the drawback of being fundamentally monochrome, and even when colorized, the accurate reproduction of colors is low. The seventh session concludes with a presentation session (I), where all students showcase their created stereoscopic VTuber anaglyph videos using a projector.

(C) Sessions eight to 11 focus on recreating the previously generated VTubers with multiple perspectives and integrating them into a multiview video for color stereoscopic viewing. For observation, students use a simplified integral photography method, setting up a lens sheet on the laptop screen, enabling autostereoscopic viewing. However, this approach has limitations in terms of the field of view available for stereoscopic viewing in a multiperson setting and potential resolution reduction when the movements are in the depth direction of the subject. During sessions 12–14, students work on their final projects. For these projects, no specific instructions or restrictions on the type of 3D content or stereoscopic techniques to be used are provided, encouraging students’ creativity and exploration. The final session, the 14th, involves all students presenting their final projects (presentation session (II)).

3 STEREOSCOPIC VTUBER USING ANAGLYPH VIDEOS

Figure 2 depicts the workflow for creating stereoscopic VTuber videos in this exercise [Hibiki-Style, 2019; Erano, 2018]. Current VTubers synchronize avatars in real-time with the speech and movements of the human operator. However, in this practical class, various types of software are applied sequentially, executing the processing step by step, thereby resulting in the offline generation of VTuber videos.

3.1 MikuMikuDance(MMD) Figure. 2 (a)

MMD is a popular and free 3D-CG animation creation tool [Higuchi-M, 2023; Saboten-P, 2020]. To use MMD, students install the 64-bit version from the MMD official website and simultaneously download and load 3D model data available on MMD and the internet. The sense of depth in the final output can be enhanced by loading background stage data.

Several default 3D models, from which students can select their avatars, are available in the pre-installed folder, such as “Hatsune Miku (Animasa)” [Crypton Future Media and INC., 2023], part of the “Piapro Characters.” Students can also use other MMD models [Menbou-shiki-Hatsune-Miku, 2022] with individual license.

Students can easily make their selected MMD 3D character dance by applying downloaded MMD motion data. parallax videos can

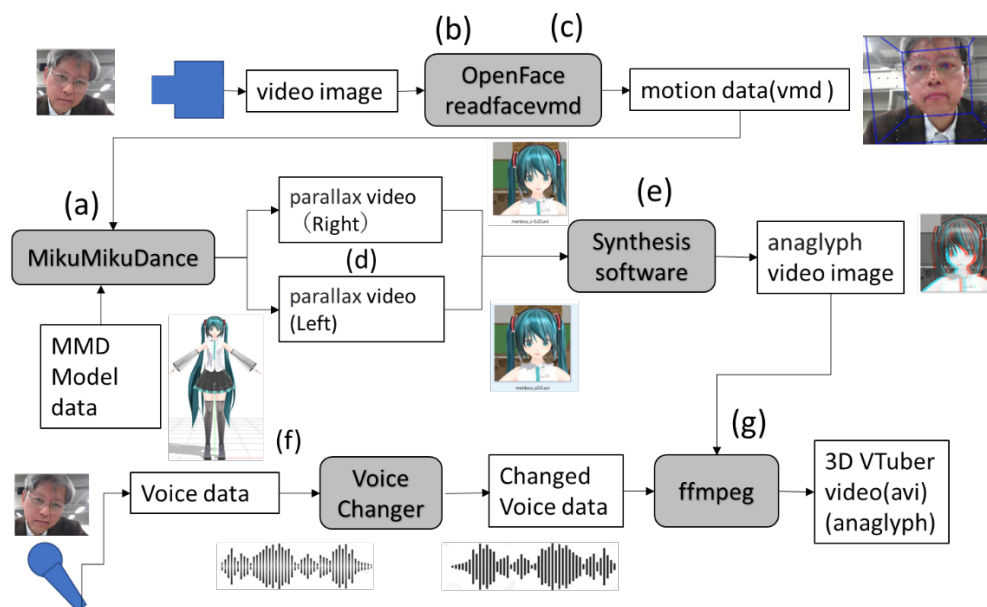


Figure 2: Flow chart of generating stereoscopic VTuber using anaglyph videos [Menbou-shiki-Hatsune-Miku, 2022].

be created and anaglyph videos can be generated by setting up a virtual camera in MMD and conducting two pseudo shootings in a 3D space. This condition allows an immersive experience of the two typical stereoscopic perceptions, “popping” and “sinking,” in the video.

3.2 OpenFace and Readfacevmd (Figures. 2 (b) and (c))

Movements of the head, limbs, eyes, and of a person should be detected, and the data are applied to move the CG avatar, thereby achieving VTubing. In the past, this operation required expensive motion capture devices that used multiple cameras or special devices, such as Time of Flight (TOF). However, in recent years, technologies utilizing artificial intelligence (AI), such as deep learning, have emerged, enabling the extraction of 3D posture and movements of individuals by using only a monocular video camera. For example, OpenPose [OpenPose, 2023] can obtain full-body skeletal information, but it requires a GPU for practical speed. The use of OpenFace [CMU_openface, 2023; Ochi-Lab, 2020; OpenFace, 2021], which operates in real-time on free software and performs light processing, even on non-GPU-equipped laptops, was preferred, considering that the data to be obtained are limited to the head. First, OpenFace is installed and verified for its operation. In addition, “readfacevmd” [Readfacevmd v1.4, 2018] requires the image processing library OpenCV 3.4.0 (Win version), which should be downloaded and installed from the Opencv.org website [OpenCV, 2023]. Readfacevmd software uses OpenFace to output motion data (vmd format) used in MMD. This software operates in an environment, where OpenFace and OpenCV 3.4.0 are installed. The motion data (vmd file) are generated by specifying a recorded video file of facial images as input. Then, MMD is used to load the avatar

3D model and background, followed by loading the motion file converted by readfacevmd.

3.3 Stereoscopic Viewing (Figures. 2. (d) and (e))

As shown in Figure. 3, only motion centered around the model’s head is effective. Thus, the video is obtained as a bust shot, capturing the model’s upper body to fill the screen.

Two main methods are used for setting up virtual cameras to create stereoscopic content, as shown in Figure. 3. With method (a), the camera is set parallel to achieve zero parallax at infinity, causing the subject to appear to “popping” in front of the PC screen. With method (b), the camera is provided a convergence angle to achieve zero parallax in front of the subject, causing the subject to appear to be “sinking” behind the PC screen.

When generating multiview videos in MMD, multiple virtual cameras cannot be set. Thus, the process involves conducting VTuber video output from MMD for each changed viewpoint. Then, anaglyph synthesis software tool provided by the instructor, is used to combine the videos to create stereoscopic viewing.

3.4 Adding Audio to VTuber Content (Figures. 2 (f) and (g))

Each student records their own voice using a voice recorder and integrates it into the video because the created 3D videos lack audio. Students record their voice with a voice recorder in m4a format using Windows Voice Recorder. They click on the recorded audio in the voice recorder, “trim” it to match the video, “save” it, and then use the “Audio Converter” [https://online-audio-converter.com/ja/.] to convert it to mp3 format. Subsequently, they use the “Voice Changer” [https://voice-changer.userlocal.jp/.], which allows freely changing the voice to any desired voice (e.g., the voice of Hatsune

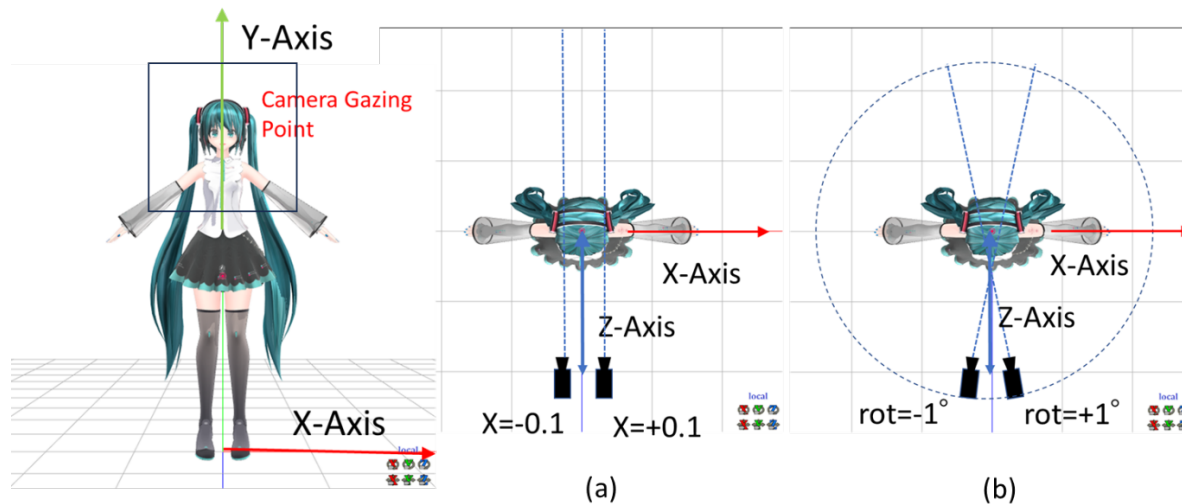


Figure 3: MMD camera settings [Menbou-shiki-Hatsune-Miku, 2022].

Miku). Finally, they use the video tool ffmpeg [<https://ffmpeg.org/>] to combine the stereo video file with the modified audio.

3.5 Presentation Session (I)

The completed audio-added anaglyph videos can be displayed using standard Windows video player software. Viewers can enjoy the videos wearing red–blue 3D glasses. 26 students’ stereoscopic VTuber works (anaglyph videos with audio) were sequentially projected onto the screen using a projector.

The submitted works had an average duration of 31.6 seconds and featured various backgrounds, with many of them mimicking the delivery style of actual VTubers, including speaking and some incorporating singing and moving backgrounds.

4 AUTOSTEREOSCOPIC VTUBER

To create autostereoscopic videos, a multiview video with typically 64 viewpoints should be generated. For this purpose, Unity [Unity Technologies, 2023], which allows multiple camera setups, is used. In Unity, various 3D models can be downloaded and used from a site called “Asset Store.” Moreover, students who have already created VTubers in MMD, can migrate their VTubers directly to the Unity environment.

4.1 MMD to Unity Model Conversion

The MMD to Unity model conversion is performed using the “MMD4Mecanim” tool [Nora@Stereoarts, 2020] for Unity; it converts Vtuber’s MMD-format (pmx/pmd) models and motion data (vmd format) to Unity format.

During this conversion, the instructions and licensing conditions of the copyright holders of the MMD models should be complied. The 3D model and motion data are stored in a single folder for conversion, consuming several minutes to over 10 minutes depending on the complexity of the model. After a successful conversion, the converted 3D model is set in Unity’s Hierarchy, and the converted

motion data are set in Unity’s Animator Controller, allowing the stereoscopic VTuber created in MMD to work in Unity in a similar manner.

4.2 Autostereoscopic viewing using Unity

After confirming the movement of the CG character converted to the Unity format, the process proceeds for stereoscopic viewing within Unity, as shown in Figure. 4. In this exercise, students place autostereoscopic display lenses, provided by the instructors, on top of the Liquid Crystal Display (LCD) of their laptops. The students create and execute a program for autostereoscopic display using Unity, following the materials provided by the instructors.

The program includes features to capture the scene with 64 virtual cameras, integrate them into a single memory area called “Render Texture,” map it to a virtual plane, convert it into an image suitable for autostereoscopic viewing using a shader, and capture the image with the main camera of Unity Editor to display the image on the PC screen.

The lens used in this exercise is a fly’s eye lens of 25.6 lines per inch (LPI) and dimensions of 150 mm (H) × 150 mm (V), using integral photography technology. This technology allows obtaining disparities in the horizontal and vertical directions, enabling stereoscopic viewing. Different from methods that require dedicated 3D glasses, such as anaglyph, which has only two views (one for the right eye and one for the left eye), the number of views in the autostereoscopic method increases because viewers must perceive stereoscopic images even when their eyes move relative to the screen. The number of views corresponds to the number of pixels on the LCD per tiny convex lens that constitute the fly’s eye lens. If this number is an integer, then it is called an integer view method. However, achieving this state is difficult unless the lenses are designed and manufactured to match the resolution of the LCD. In this exercise, the LCD on the students’ laptops and the autostereoscopic lenses used are independently designed and

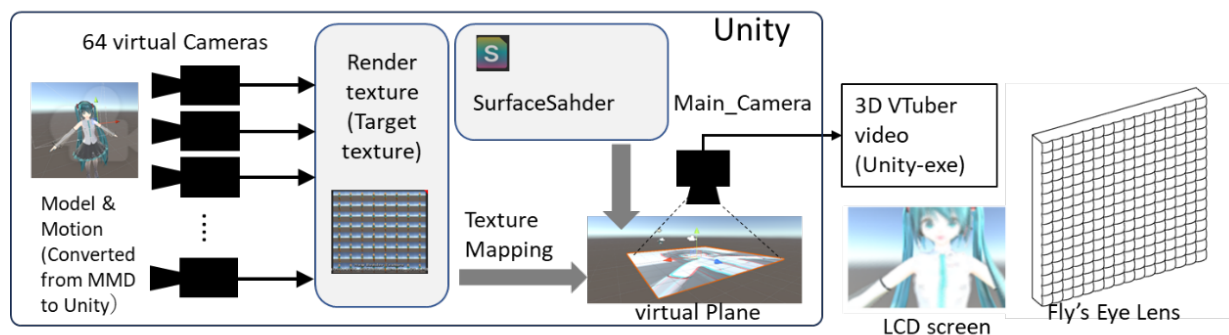


Figure 4: Autostereoscopic viewing using Unity [Menbou-shiki-Hatsune-Miku, 2022].

manufactured. Thus, the number of views becoming an integer is highly unlikely; typically, it will be a non-integer.

Therefore, the “Extended Fractional View” method proposed by Yanaka, which is an extension of Ishii’s fractional view method for lenticular lenses, is used in conjunction with integral photography [Ishi, 2004; Yanaka, 2007].

4.3 Presentation Session (II)

A total of 25 student’s works were received, where 16 of them being anaglyph videos and nine being autostereoscopic videos. Among the 16 anaglyph videos, the content consisted of 3D-CG (MMD) models combined with downloaded dance motions from the internet. During the presentation, the autostereoscopic works were displayed on individual PCs for each participant to observe and evaluate in three stages, with everyone taking turns in front of their PCs. In addition, the anaglyph videos were projected for simultaneous evaluation, similar to that in the previous presentation session (I).

5 CLASS EVALUATION

5.1 Interest and Difficulty in the Class

At the end of the course, a survey was conducted among the 25 participating students who attended the final class. The results are shown in Figures 5 (a)–(d). The overall impression received a high rating with 88% responding as “good” or “excellent,” indicating that the majority of students found the class engaging. In terms of difficulty of the courses, 52% responded with “hard” or “very hard,” signifying that half of the students felt that it was challenging. In the final presentation, 76% of the students provided affirmative answers, ranging from “good” to “excellent.” Moreover, 72% of the students expressed the desire to “apply what they learned in the future.”

Individual comments included the following:

“As opportunities to view models in stereoscopically are rare, I felt that being able to animate models with parallax was a very valuable experience.”

“Rare encounter of combining recent VTuber’s face recognition system and stereoscopic function provided a good experience.”

“Overall, I was able to learn about interesting subjects. Especially, obtaining facial information with OpenFace for VTuber was very fascinating. I thought that applying this approach to Unity and other platforms would be fun.”

“It was my first time loading MMD into Unity, and I learned how to import models and motions, which was a great experience. The Integral method required dedicated lenses, but it left a strong impression as it looked great even without glasses.”

5.2 3D Content and Stereoscopic Methods

In Figure 5(e), when asked about the 3D content that sparked the most creative interest, 48% of the students preferred stereoscopic content created by downloading existing MMD models and motions. In fact, all anaglyph video works at the presentation session (II) were made in this format. This result indicates that the students felt a new sense of value in enabling their favorite 3D CG character to dance with high-quality motions and then view it in stereoscopic 3D. In addition, 40% of the students were motivated to create stereoscopic VTubers by combining existing MMD models with motion capture. VTuber content generally requires special talents and skills, such as entertaining talk, game commentary, and singing; thus, creating these works can be challenging. Nevertheless, 40% of the students were most eager to explore this approach, demonstrating the potential of using stereoscopic VTubers in educational contents.

In Figure 5(f), when asked about the most intriguing stereoscopic method, 48% preferred anaglyph, 40% chose lenticular lens and fly’s eye lens for autostereoscopic viewing, and 12% selected polarized viewing. For the final projects, 64% chose anaglyph videos and 36% chose autostereoscopic work, with both numbers being roughly identical to the survey results. Thus, students seem to have experienced the technical difficulty of autostereoscopic methods; individual comments included the following:

“Adjusting autostereoscopic viewing was difficult because it had to be adjusted based on the screen.”

“When using lenticular or fly’s eye lens methods, adjusting Unity parameters (to prevent blurriness) was

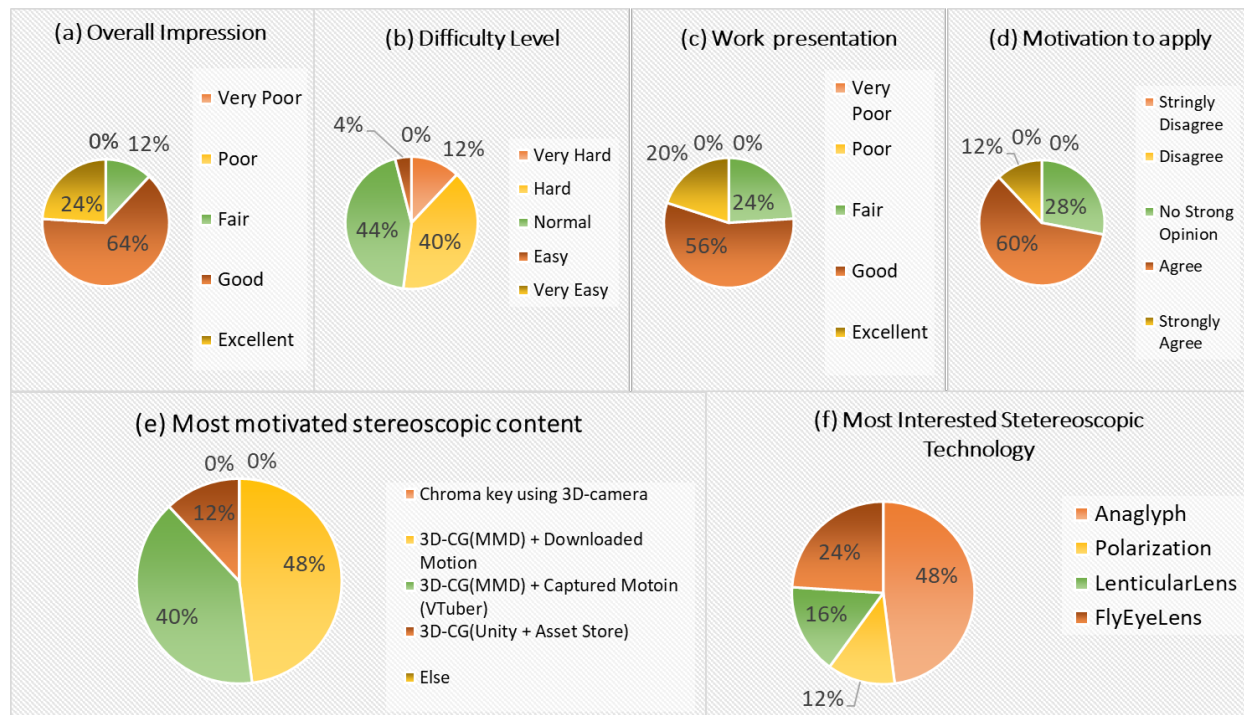


Figure 5: Results of student survey.

challenging, but the sense of accomplishment when it worked well was satisfying.”

6 CONCLUSIONS

In this study, we generated VTuber videos that can be viewed in stereoscopic vision using motion extracted from monocular images. With the use of freely available tools and the students’ own laptops, combined with the red–blue glasses, lenticular lenses, and fly’s eye lenses, remarkable results could be achieved. Incorporating popular contents from the internet into the exercise contributed to the high praise received from the students. However, because the process involved offline video production without real-time capabilities, it may not be suitable for the current mainstream real-time VTuber usage. Nonetheless, we hope that many students’ interest in this field will serve as a catalyst for further application and development of this technology. The exercise achieved a high level of educational content by reproducing the same VTuber CG content using different stereoscopic display methods, namely, anaglyph, lenticular, and fly’s eye lens displays. Furthermore, it allowed students to explore system-oriented thinking techniques by combining various tools that had been rarely considered in university education. As a result, this practical course holds significant values as an education that combines information media art, system-oriented thinking, and AI technologies, such as OpenFace.

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